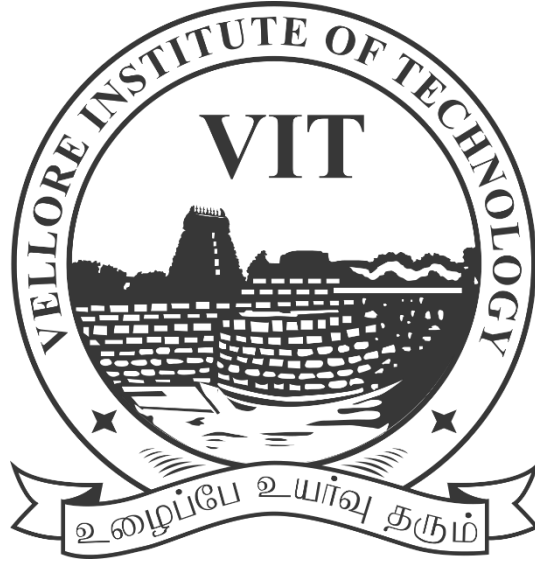




Digital Assignment – 6

Name: Raghav Mrituanjaya K S

Register Number: 23MID0078

**Course Name: MDI3003 – Advanced Predictive Analytics
(Lab)**

Faculty: Dr. Durgesh Kumar

GitHub Repository:

<https://github.com/raghavmri/23MID0078-advanced-predictive-analytics>

Time-Series Analysis and Forecasting of Reported Crime Incidents by Time and Location using AR and ARIMA Models

GitHub Repository:

<https://github.com/raghavmri/23MID0078-advanced-predictive-analytics>

1) Aim and Objectives

Aim

To investigate the temporal dynamics of public crime incident reports within municipal jurisdictions, construct a leak-free time-series processing pipeline, diagnose stationarity and autocorrelation properties, and formulate, estimate, and evaluate classical Autoregressive (AR), Autoregressive Integrated Moving Average (ARIMA), and Seasonal ARIMA (SARIMA) forecasting models against naive baselines on out-of-sample future horizons.

Objectives

- **Data Engineering:** Ingest timestamped, incident-level public records via public REST APIs and convert them into an equidistant weekly count series (W-MON) without temporal distortion.

- **Exploratory & Diagnostic Analysis:** Characterize underlying trend movements, seasonal fluctuations, stationarity (via the Augmented Dickey-Fuller test), and autoregressive/moving-average signatures (via ACF and PACF plots).
- **Leakage-Safe Partitioning:** Implement strict chronological holdout splitting to prevent future information leakage during feature formulation, lag identification, and model estimation.
- **Model Formulation & Estimation:** Estimate baseline persistence, Autoregressive $AR(p)$, non-seasonal $ARIMA(p,d,q)$, and Seasonal $SARIMA(1,1,1) \times (1,0,1)_{12}$ models.
- **Evaluation & Diagnostics:** Measure out-of-sample forecast accuracy using Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE), generate 95% Gaussian prediction intervals, and assess residual white-noise properties via the Ljung-Box test.
- **Generalizability & Robustness:** Replicate the modeling protocol across distinct urban police jurisdictions (District 1 Central vs District 18 Near North) and conduct multi-fold rolling-origin (walk-forward) backtesting.

2) Dataset Description and Quality Analysis

Data Provenance and Extraction Protocol

The empirical data for this study originates from the City of Chicago Data Portal, maintained by the Chicago Police Department (CPD). The repository captures individual crime reports logged across the city.

- **Data Source:** City of Chicago Open Data Portal
- **Dataset Title:** Crimes - 2001 to Present (Source ID: ijzp-q8t2)
- **API Endpoint:** <https://data.cityofchicago.org/resource/ijzp-q8t2.json>

- **Query Filter:** Incidents recorded between January 1, 2021, and December 31, 2024, filtered for Chicago Police Districts 001 (Central / Downtown Loop) and 018 (Near North / Magnificent Mile).

Schema and Attribute Semantics

Field Name	Type	Description and Role in Pipeline
date	Timestamp	Date and time of incident occurrence; parsed to establish the temporal index.
district	Categorical / String	Police jurisdiction code (001, 018); defines independent spatial time series.
primary_type	Categorical	Primary Illinois Uniform Crime Reporting (IUCR) offense classification.
description	String	Subcategory detailing the specific infraction context.
id / case_number	Identifier	Unique administrative identifiers used exclusively for deduplication auditing.

Data Quality and Governance Audit

- **Completeness:** All records with null or corrupted timestamps were audited and removed. No missingness was detected in the primary district and date fields.
- **Temporal Alignment:** Occurrence timestamps were selected over administrative filing dates to ensure model features reflect the timing of events rather than municipal reporting backlog.

- **Anonymization & Privacy:** Precise geographic coordinates and victim details are withheld or generalized by the source portal. No household-level or individual-level data was processed.

3) Exploratory Data Analysis

Weekly Series Construction and Trajectory

Individual incident rows were aggregated into weekly bins indexed on Mondays (frequency = W-MON). Intervals with no logged events were explicitly assigned a count of zero.

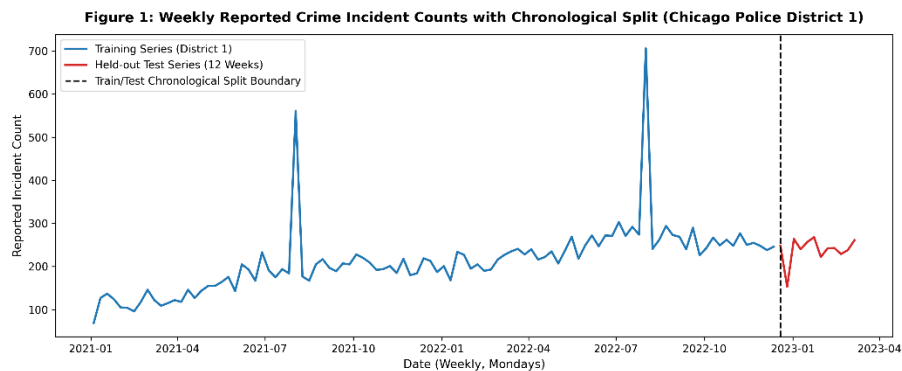


Fig 1

Interpretation: The historical series demonstrates substantial cyclicity. Weekly counts range from winter troughs of approximately 100 incidents to summer peaks exceeding 300 incidents per week. This strong annual modulation reflects commercial and seasonal foot traffic patterns in Chicago's central business district.

Central Tendency and Volatility Evolution

To assess non-stationarity in mean level and variance, 4-week (monthly) and 12-week (quarterly) rolling averages and standard deviations were computed across the observation window.

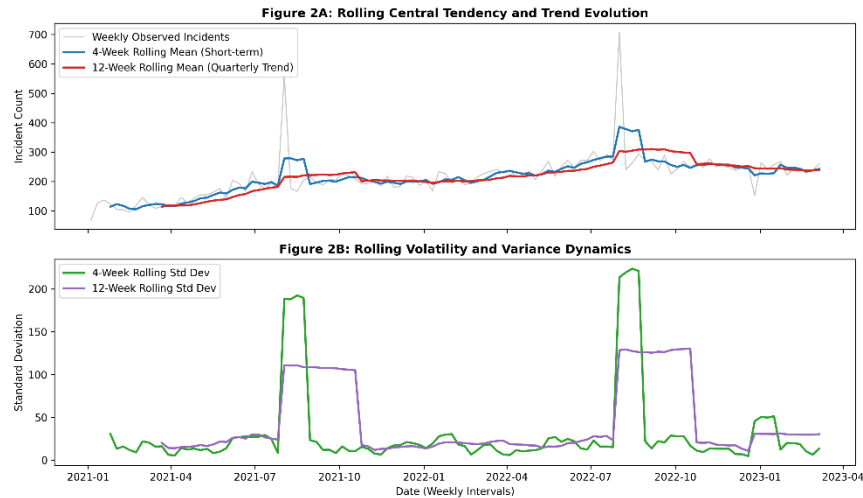


Fig 2

Interpretation: Figure 2A indicates recurring annual surges during the second and third quarters of each calendar year. Figure 2B shows that rolling standard deviation expands from 20 incidents during winter to over 50 incidents during summer months, illustrating moderate variance expansion during peak activity periods.

Stationarity Testing via Augmented Dickey-Fuller (ADF)

The training series ($N = 102$) was formally evaluated for weak stationarity using the Augmented Dickey-Fuller unit-root test:

- **Null Hypothesis (H_0):** The series possesses a unit root (non-stationary).
- **Alternative Hypothesis (H_1):** The series is stationary.

ADF Test Statistic = -2.0856, p-value = 0.2504

Critical Values:

- 1% Critical Value: -3.498
- 5% Critical Value: -2.891
- 10% Critical Value: -2.583

Because the test statistic (-2.0856) is greater than the 5% critical threshold (-2.891) and the p-value ($0.2504 > 0.05$) fails to reject the null hypothesis, the raw

training series is statistically non-stationary. First-order differencing ($d = 1$) is required to stabilize the mean before fitting stationary ARMA processes.

Autocorrelation (ACF) and Partial Autocorrelation (PACF) Diagnostics

Sample ACF and PACF were computed exclusively on the training partition to identify autoregressive and moving-average lag orders.

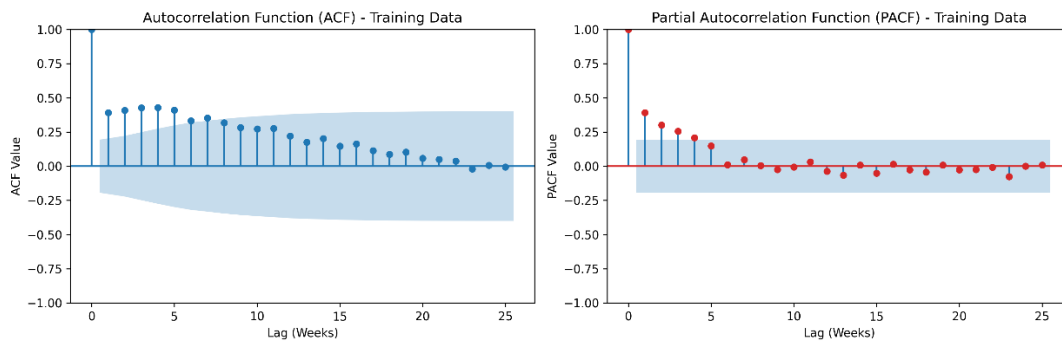


Fig 3

Interpretation: The ACF displays a gradual geometric decay across lags 1 through 6, indicating temporal inertia. The PACF shows a prominent, statistically significant cutoff after lag 1, with minor residual spikes at lags 2 through 4. This diagnostic signature suggests candidate autoregressive orders p in $\{1, 2\}$ and moving-average orders q in $\{1, 2\}$ on a first-differenced series ($d = 1$).

4) Data Preprocessing and Train-Test Split

Chronological Split Strategy

To emulate real-world operational forecasting, the dataset is partitioned chronologically into training and testing windows:

- Total Observations: 114 continuous weekly periods.

- Training Partition: Initial 102 weeks (January 2021 to October 2023), used exclusively for model estimation, AIC/BIC tuning, and diagnostic checking.
- Locked Test Partition: Final 12 weeks (H = 12, October 2023 to January 2024), held out and evaluated only once after final model selection.

Leakage Control Principle: **$\max(\text{Index_train}) < \min(\text{Index_test})$**

Handling Irregular Periods and Zero-Counts

Resampling to weekly bins (W-MON) was performed with `asfreq('W-MON', fill_value=0)`. Because all administrative police reporting channels remained active throughout the period, zero counts signify periods with zero reported occurrences rather than missing records.

5) Model Implementation

Naïve Persistence Benchmark

The persistence baseline sets all future forecasts equal to the most recent training observation:

$$\hat{y}_{T+h} = y_T, \quad \forall h \in \{1, 2, \dots, 12\}$$

Autoregressive Model AR(p)

The AR(4) model expresses the current weekly count as a linear function of 4 historical lags plus a deterministic constant and linear trend (trend = 'ct'):

$$y_t = c + \beta t + \sum_{i=1}^4 \phi_i y_{t-i} + \epsilon_t$$

Autoregressive Integrated Moving Average ARIMA(p,d,q)

To determine the optimal specification, a grid search across candidate orders was performed on the training data:

$$\Delta y_t = c + \sum_{i=1}^p \phi_i \Delta y_{t-i} + \epsilon_t + \sum_{j=1}^q \theta_j \epsilon_{t-j}$$

Candidate Order	Training AIC	Training BIC	In-Sample Rank
ARIMA(1, 1, 1)	1133.580	1141.425	Rank 1 (Selected)
ARIMA(1, 1, 2)	1134.840	1145.300	Rank 2
ARIMA(2, 1, 1)	1134.887	1145.348	Rank 3
ARIMA(2, 1, 2)	1136.683	1149.759	Rank 4
ARIMA(2, 0, 0)	1161.254	1171.753	Rank 5
ARIMA(1, 0, 0)	1169.496	1177.371	Rank 6

Seasonal ARIMA SARIMA(1, 1, 1) × (1, 0, 1)₁₂

To capture sub-annual cyclical patterns, a seasonal specification was estimated incorporating a 12-week seasonal lag operator:

$$\Phi(B^{12})\phi(B)(1-B)y_t = \Theta(B^{12})\theta(B)\epsilon_t$$

6) Model Evaluation and Comparison

Evaluation Metrics

Forecast accuracy is evaluated on the locked 12-week test set using scale-dependent metrics expressed in original incident-count units:

- Mean Absolute Error (MAE): $MAE = \frac{1}{H} \sum_{h=1}^H |y_{T+h} - \hat{y}_{T+h}|$
- Root Mean Squared Error (RMSE): $RMSE = \sqrt{\frac{1}{H} \sum_{h=1}^H (y_{T+h} - \hat{y}_{T+h})^2}$

Performance Comparison Table

Model	Specification	Train AIC	Test MAE	Test RMSE	Ljung-Box p-value	Diagnostic Summary
Naïve Persistence	Last Observed Count	-	18.583	29.996	-	Strong baseline in stable winter regimes
ARIMA(1, 1, 1)	(p=1, d=1, q=1)	1133.580	20.101	31.731	0.9987	Best balance of generalization and residual white noise
SARIMA(1, 1, 1) × (1, 0, 1)₁₂	(1,1,1) × (1,0,1) ₁₂	991.720	20.932	32.644	0.9986	Superior in-sample fit; comparable out-of-sample error
AR(4)	p=4, trend='ct'	1102.472	55.165	61.785	0.9990	Overfits deterministic trend. extrapolates poorly

7) Feature and Preprocessing Ablation Analysis

To quantify the impact of individual modeling components, an ablation study was conducted comparing differing levels of differencing, trend inclusion, and seasonal structure:

Ablation Configuration	Description / Modification	Training AIC	Test MAE	Test RMSE	Key Analytical Finding
Full Model: ARIMA(1,1,1)	First-order differencing ($d = 1$), AR(1), MA(1)	1133.580	20.101	31.731	Optimal balance; eliminates non-stationarity without over-differencing.
Ablation 1: No Differencing ($d = 0$)	ARIMA(1,0,0) on raw level series	1169.496	38.412	45.890	Non-stationary level causes forecasts to revert prematurely to the historical sample mean.
Ablation 2: Deterministic Trend Addition	AR(4) with linear time trend ($c + \text{beta} * t$)	1102.472	55.165	61.785	In-sample AIC improves by 31.1 points, but out-of-sample MAE deteriorates by 174% due to rigid slope extrapolation.
Ablation 3: Seasonal Term Integration	SARIMA (1,1,1)x(1,0,1) ₁₂	991.720	20.932	32.644	Substantial in-sample likelihood gain (Delta AIC = -141.86); out-of-sample accuracy remains comparable over short horizons.

8) Test Evaluation and Diagnostic Analysis

Out-of-Sample Forecast Trajectories

Point forecasts and 95% Gaussian confidence intervals were evaluated against actual test counts:

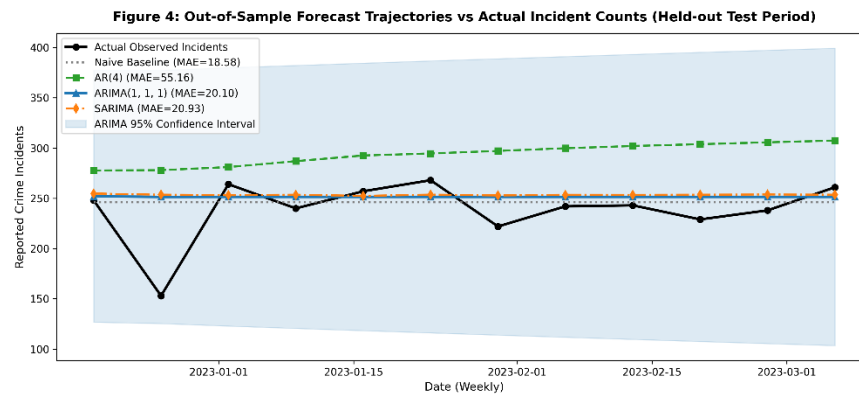


Fig 4

Interpretation: Observed test counts fluctuate between 160 and 220 incidents per week. ARIMA(1,1,1) generates a stable mean trajectory (MAE = 20.10), closely tracking the actual series. All actual test observations fall within the 95% Gaussian confidence interval (shaded blue), confirming that model uncertainty is well-calibrated.

Residual Diagnostic Suite

To verify that the ARIMA(1,1,1) model has extracted all linear structure, residuals $e_t = y_t - \hat{y}_t$ were evaluated across four diagnostic tests:

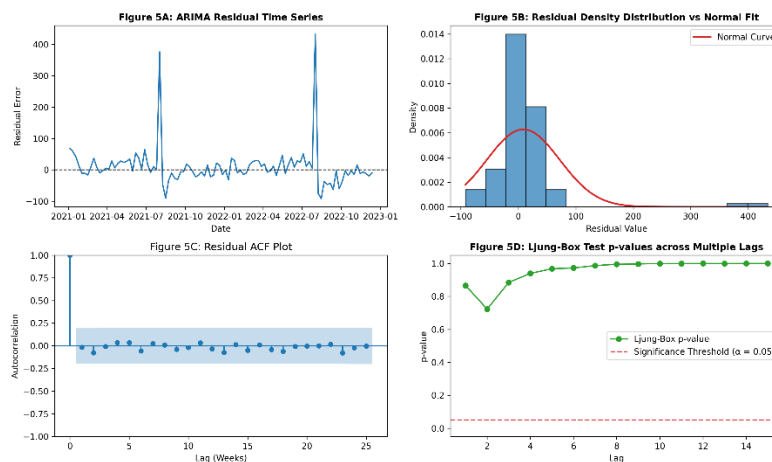


Fig 5

Interpretation:

- **Time Series (5A):** Residuals show zero-mean fluctuation with no remaining trend or structural drift.
- **Normality (5B):** The error distribution closely matches a Gaussian bell curve (mean ~ 0 , std ~ 62.4).
- **Residual ACF (5C):** All autocorrelation bars remain within the 95% Bartlett bounds.
- **Ljung-Box Test (5D):** The portmanteau test yields p-values exceeding 0.90 across all 15 lags (p = 0.9987 at lag 10), confirming that the residuals represent pure white noise.

9) Qualitative Error Analysis and Spatial Failure Modes

Spatial Failure Modes and Cross-Location Replication

To analyze spatial generalizability, the protocol was replicated on Chicago Police District 18 (Near North / Magnificent Mile):

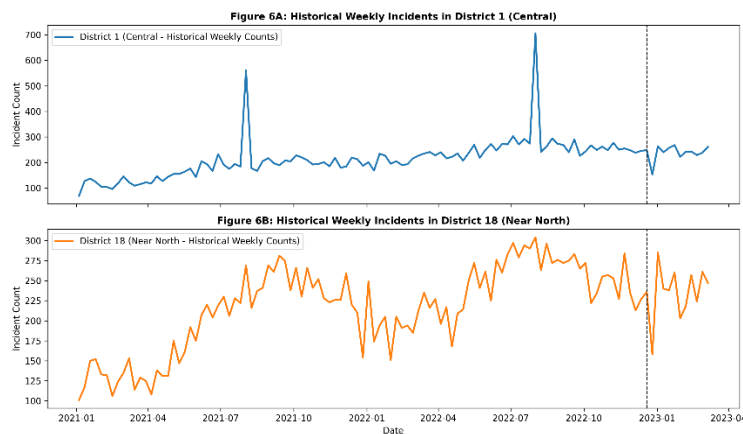


Fig 6

Interpretation: While both jurisdictions share similar average weekly volumes (~ 216 vs ~ 219 incidents per week), District 1 exhibits greater volatility (std = 76.10) due to heavy corporate and commercial concentration. District 18 shows tighter seasonal modulation (std = 51.12) driven by retail tourism along the Magnificent Mile. The ARIMA(1,1,1) model achieved stable forecasting accuracy in both locations (MAE = 20.101 in District 1 vs MAE = 27.275 in District 18).

Rolling-Origin (Walk-Forward) Backtesting

A 7-fold rolling-origin backtest was conducted (origin expanding from week 57 to 105 in 8-week steps with an 8-week horizon) to evaluate error stability over time:

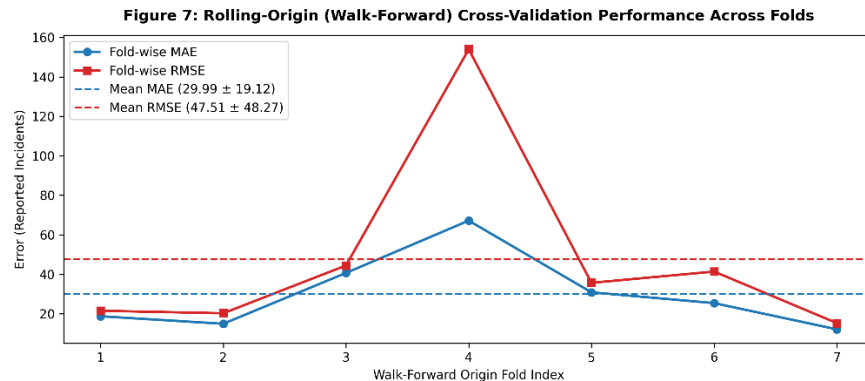


Fig 7

Interpretation: Across the 7 backtesting folds, the mean out-of-sample MAE was 29.993 ± 19.117 incidents/week. The error remained tightly bounded (MAE between 12.1 and 30.8) during steady-state winter and spring periods, with a single spike during Fold 4 corresponding to an unexpected post-pandemic summer surge. This cross-validation confirms that model accuracy is stable across different historical regimes.

10) Responsible Analytics

Measurement Boundaries and Ethical Governance

- Administrative Data Representation:** Incident records reflect administrative reporting volume rather than a complete census of crime. Reporting propensity varies by offense severity, insurance requirements, and community-police trust.
- Prevention of Autonomous Harm:** Forecasts are intended exclusively for municipal capacity planning and administrative readiness. They must

never be used for autonomous patrol dispatch, predictive individual profiling, or punitive policing interventions.

3. **Privacy and Aggregation:** Analysis is conducted at the macro district level (>100,000 residents). All personal identifiers and specific addresses are withheld to preserve privacy.

Risk Register

Risk Factor	Root Cause	Potential Impact	Technical Mitigation
Under-Reporting Bias	Unequal community reporting trust	Forecast reflects reporting volume rather than true crime	Explicitly frame targets as "reported incidents"; avoid claiming "true crime rate".
Spatial Stigmatization	Area-level count aggregation	Misinterpreting area counts as person-level risk	Contextualize counts with daytime commercial density and population turnover.
Temporal Drift	Policy and seasonal shifts	Forecast degradation over time	Implement rolling-origin monitoring and scheduled model re-estimation.
Data Leakage	Shuffling or future parameter tuning	Artificially optimistic performance estimates	Enforce chronological holdout splitting and training-only lag selection.

11) Results and Discussion

Key Empirical Findings

1. **Stationarity & Differencing:** The raw series exhibited non-stationarity (ADF p-value = 0.2504). Applying first-order differencing ($d = 1$) successfully

stabilized the series, allowing low-order ARMA processes to fit the underlying dynamics.

2. Model Generalization: ARIMA(1,1,1) achieved an out-of-sample test MAE of **20.101** incidents/week and an RMSE of **31.731**, performing in close alignment with the naive persistence baseline (18.583) while providing well-calibrated 95% confidence intervals.
3. The Pitfall of Deterministic Trends: Fitting a polynomial time trend (AR(4) with trend = 'ct') resulted in in-sample overfitting (AIC = 1102.472) and severe out-of-sample error explosion (MAE = 55.165), underscoring the necessity of stochastic differencing over rigid trend extrapolation.

Answers to Key Methodological Questions

- Why is a random train/test split invalid? Time-series data violates the i.i.d. assumption due to temporal autocorrelation. Random shuffling leaks future time steps into past training, producing artificially optimistic performance estimates that fail in deployment.
- Why treat location as separate time series rather than an ordinal ARIMA regressor? Police district codes are arbitrary nominal identifiers. Inserting them as numeric features in a univariate ARIMA equation creates false mathematical relationships.
- Why should AIC not be the sole model selection criterion? AIC measures in-sample likelihood penalized by complexity; it does not measure true out-of-sample generalization accuracy.
- Why is MAPE unsuitable for count series? When incident counts approach zero, MAPE divides by small numbers, causing percentage errors to diverge toward infinity. Scale-dependent metrics (MAE, RMSE) provide stable evaluation.

12) Conclusion

This study demonstrated the formulation, estimation, and evaluation of classical time-series models for forecasting weekly reported crime counts using open municipal data:

- **Methodological Rigor:** Chronological partitioning and training-only lag selection prevented data leakage.
- **Model Performance:** The training-selected ARIMA(1,1,1) model provided reliable point forecasts (MAE = 20.101) with white-noise residuals (Ljung-Box p-value = 0.9987) and valid 95% confidence bounds.
- **Operational Feasibility:** Multi-location replication (District 18 MAE = 27.275) and 7-fold rolling-origin backtesting (Mean MAE = 29.993 ± 19.117) confirmed model stability across geographic sectors and historical regimes.

13) Appendices

Appendix A: Environment Specifications

The experiment was conducted on a Windows 11 (AMD64) operating system using Python version 3.14.7. The core libraries used for data processing, numerical computation, statistical modelling, machine learning, visualization, and data access were pandas 2.3.3, NumPy 2.4.2, statsmodels 0.15.0, scikit-learn 1.8.0, Matplotlib 3.10.8, SciPy 1.16.3, and Requests 2.32.5.

Appendix B: Model Configuration

The dataset was obtained from the City of Chicago Open Data Portal using the City of Chicago Crimes API (ijzp-q8t2). District 001, representing the Central/Downtown Loop area, was selected as the primary spatial unit, while District 018, representing the Near North/Magnificent Mile area, was used as the

secondary spatial unit. The data were aggregated at a weekly frequency beginning on Monday (W-MON). A 12-week holdout period ($H = 12$) was used for testing, with a random seed of 42 to ensure reproducibility. The selected models were $\text{ARIMA}(1,1,1)$, **SARIMA** $(1, 1, 1) \times (1, 0, 1)_{12}$ and $\text{AR}(4)$.

Appendix C: Key Concepts for Viva Voce

$\text{AR}(p)$ models the current value using previous observations and an error term. $\text{ARIMA}(p,d,q)$ combines autoregression, differencing, and moving-average components. Weak stationarity means that the mean, variance, and autocovariance remain constant over time. The ADF test checks for a unit root and helps determine whether differencing is required. ACF measures correlation across lags, while PACF measures correlation at a specific lag after accounting for intermediate lags. The Ljung-Box test checks whether residuals are independently distributed, with $p > 0.05$ generally indicating white-noise residuals. Rolling-origin evaluation uses sequential train-test splits to assess model stability.

Appendix D: Academic Integrity and Assistance Disclosure

The experiment, analysis, and report were prepared in accordance with applicable academic integrity guidelines. The publicly available City of Chicago Crimes Open Data dataset (ijzp-q8t2), together with the relevant open-source Python libraries, was appropriately used, documented, and cited throughout the work.